Supplementary Tables for

Effective Archimedean Stark Theorems over Real Quadratic Fields

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Scope and archive

This supplement contains the record map and the complete Artin-label interval replay referenced by the main paper. It also records queue-level Engine-A statistics that are not used in the uniform theorem.

Archive DOI. `https://doi.org/10.5281/zenodo.21707692`. The main paper PDF and TeX source, this supplement, and the frozen companion archive are top-level files in the public record.

Supplementary Table S1: certificate record map

result	record
first order six	`data/q7-p7-case-v1.json`
second order six	`data/q14-p7-case-v1.json`
RQ-000108	`data/rq000108-case-v1.json`
RQ-000021	`data/rq000021-case-v1.json`
ramified-prime-3 control	`data/q57-norm27-case-v1.json`
RQ-002955	`data/rq002955-case-v1.json`
order ten	`data/q33-p11-order10-case-v1.json`
cyclic-quartic character selection	`artifacts/engine-c-character-selection-v1.json`
$\mathbb{Q}(\sqrt{35})$ CM packet	`artifacts/engine-c-w3-tranche-01-verified-v1.json`
$e = 6$ powered-unit record	`artifacts/engine-c-e6-tranche-01-verified-v1.json`
$e = 6$ primitive correction	`artifacts/engine-c-e6-primitive-packet-correction-v1.json`
$\mathbb{Q}(\sqrt{6})$, $e = 8$ proof route	`data/q6-norm8-case-v3.json`
Engine-C scope correction	`artifacts/engine-c-claim-scope-correction-v1.json`
Engine-C Fourier-sign correction	`artifacts/engine-c-fourier-convention-correction-v1.json`
dual-route packet	`data/rq000458-dual-case-v1.json`
all archimedean places	`artifacts/engine-b-archimedean-place-audit-v1.json`
quadratic theorem	`data/engine-a-uniform-theorem-v1.json`
quadratic Euler degeneracy audit	`artifacts/engine-a-euler-degeneracy-v1.json`
Shintani source map	`artifacts/shintani-1978-source-map-v1.json`
index-parity theorem	`artifacts/results-paper-index-parity-lemma-v1.json`
parity audit	`artifacts/results-paper-odd-index-parity-audit-v1.json`
literature perimeter	`data/literature-perimeter-v1.json`

result	record
Roblot sextic-overlap audit	`artifacts/roblot-sextic-overlap-audit-v1.json`

Supplementary Table S3: relation to Roblot's sextic theorem

Here H/K is the cyclic sextic one-place ray field. The columns record Roblot's hypotheses (A1)–(A3), the certified class number of H, and applicability of [1, Thm. 7.1]. Every class number was computed with `bnfinit(P,1)` and certified by `bnfcertify(bnf)=1`; the local assertion was checked in H/H^+ by exact prime decomposition. Roblot's (A4) is not part of the existence statement in Theorem 7.1; as his footnote to (P2) explains, (A4) is needed instead to deduce (P2) from a conjecturally given Stark unit. It is therefore neither assumed nor silently included in the applicability column below.

case	A1	A2	A3	h_H	Thm. 7.1
RQ-000190	yes	yes	yes	1	applies
RQ-000419	yes	yes	yes	1	applies
RQ-000021	yes	yes	yes	1	applies
RQ-002057	yes	yes	yes	1	no: wild at 3
RQ-002955	yes	yes	yes	1	applies

For RQ-002057 the relative ramification index above 3 is 6. For the other four cases no prime above 3 ramifies in H/K.

Supplementary Table S2: complete Artin-label interval replay

For each cyclic ray group below, r denotes the power of the generator in the pinned PARI ray coordinates. Each rational interval contains exactly one positive root of the absolute packet polynomial.

case	r : isolating interval
RQ-000190	$0:(5.325, 5.326)$, $1:(2.713, 2.714)$, $2:(2.251, 2.252)$, $3:(.187, .188)$, $4:(.368, .369)$, $5:(.444, .445)$
RQ-000419	$0:(20.431, 20.432)$, $1:(.189, .190)$, $2:(1.090, 1.091)$, $3:(.048, .049)$, $4:(5.289, 5.290)$, $5:(.916, .917)$
RQ-000108	$0:(7.892, 7.893)$, $1:(2.242, 2.243)$, $2:(.126, .127)$, $3:(.445, .446)$
RQ-000021	$0:(6.109, 6.110)$, $1:(7.496, 7.497)$, $2:(4.351, 4.352)$, $3:(.163, .164)$, $4:(.133, .134)$, $5:(.229, .230)$
RQ-002057	$0:(5.709, 5.710)$, $1:(.028, .029)$, $2:(.035, .036)$, $3:(.175, .176)$, $4:(34.732, 34.733)$, $5:(27.792, 27.793)$
RQ-002955	$0:(18.232, 18.233)$, $1:(2.180, 2.181)$, $2:(8.620, 8.621)$, $3:(.054, .055)$, $4:(.458, .459)$, $5:(.116, .117)$
RQ-001107	$0:(8.645, 8.646)$, $1:(.276, .277)$, $2:(1.245, 1.246)$, $3:(.571, .572)$, $4:(.236, .237)$, $5:(.115, .116)$, $6:(3.611, 3.612)$, $7:(.802, .803)$, $8:(1.748, 1.749)$, $9:(4.234, 4.235)$

Engine-A queue statistics

The uniform theorem does not depend on a finite search range. In the frozen verification queue there are 1,560 rows with nonempty differenced character support, comprising 2,232 supported quadratic character occurrences across 912 distinct character fields. Exact imprimitive Euler-factor evaluation finds 672 zero Euler products, affecting 603 rows. In 346 rows every supported derivative vanishes, so the theorem's product is empty and $X_A = 1$.

References

[1] X.-F. Roblot, *Index formulae for Stark units and their solutions*, Pacific J. Math. **266** (2013), 391–422, `https://doi.org/10.2140/pjm.2013.266.391`.